Savik Kinger

PhD Candidate in NeuroAI

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Education

2024–Present **PhD in Computer Science**, *Yale University*, New Haven, CT
Focus on NeuroAI, AI Interpretability. I develop new approaches to infer and learn the mechanisms underlying circuit dynamics in the Brain and LLMs. Adviser: Steven Zucker. Expected graduation: May 2028.

2020–2024 **BA Computer Science, BA Mathematics**, *Columbia University*, New York, NY
Columbia College, Graduated Cum Laude, GPA: 3.95

Relevant Experience

2024–Present **Graduate Research Assistant**, *Zucker Group*, New Haven, CT
Researching approaches to understanding network dynamics in AI and neuroscience by building generative models and leveraging tools from differential geometry and causal inference.

2022–2024 **Mathematical Biology Researcher**, *Rabadan Lab, Columbia University*, New York, NY
Utilized mathematical tools from AI, Topology and Optimal Transport to work on problems in Genomics.

2022–2023 **Product Research and Management Intern**, *MathWorks*, Natick, MA
Built and tested proof-of-concept Deep Learning Transformer model to assess viability for a Deep Learning-based product offering. Researched and implemented a new algorithm for MATLAB Set Difference in C++, leading to substantial performance improvements for hundreds of MATLAB functions.

Publications

[1] **S. Kinger**, J. Bertram, L. Dyballa, T.A. Kellar and S. Zucker, "Reading Positional Coupling in Transformers with Diffusion Scores," *Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026)*, Sydney, Australia. Poster, Main track. https://neurips.cc/virtual/2026/loc/atlanta/poster/155989

[2] **S. Kinger**, J. Bertram, L. Dyballa, E. Yemini and S. Zucker, "Inferring Active Neural Circuits Using Diffusion Scores," *90th Cold Spring Harbor Laboratory Symposium on Quantitative Biology, May 2026*, Cold Spring Harbor, New York, poster. https://arxiv.org/abs/2605.02852

[3] J. Bertram, L. Dyballa, T. A. Keller, **S. Kinger** and S. Zucker "Decoding Alignment without Encoding Alignment: A critique of similarity analysis in neuroscience," *Mechanistic Interpretability Workshop at the 43rd International Conference on Machine Learning (ICML 2026)*, Seoul, South Korea. poster. https://arxiv.org/abs/2605.05907

[4] J. Bertram, L. Dyballa, T. A. Keller, **S. Kinger** and S. Zucker,"Manifolds And Modules: How Function Develops In a Neural Foundation Model," *Findings of the Workshop on Data on the Brain & Mind (DBM 2025) at NeurIPS 2025*, San Diego, CA, USA, poster. https://arxiv.org/abs/2512.07869v1

[5] **S. Kinger**, "Quantifying Recombination Bias in Viral Evolution Using Persistent Homology," *Proceedings of ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB 2023)*, ACM, New York, NY, USA. https://doi.org/10.1145/3584371.3612992.

In Submission

[S1] **S. Kinger**, L. Dyballa and S. W. Zucker, "The Limits of Clustering: Geometric Thresholds Under the Manifold Hypothesis," *AISTATS 2027*. In submission. https://arxiv.org/abs/2609.17892

[S2] **S. Kinger**, J. Bertram, L. Dyballa, E. Yemini and S. W. Zucker, "Diffusion model infers lag-specific directed interactions in sampled neural data," *COSYNE 2027*. In submission.

[S3] J. Bertram, L. Dyballa, T. A. Keller, **S. Kinger** and S. W. Zucker, "Evaluating Large Foundation Models Like a Neurophysiologist," *Fabric Workshop at WACV 2027*. In submission.

Fellowships

2024-2025 Nathan Hale Fellowship, Yale University

Invited Talks and Workshops

March 2026 Indian Institute of Technology (IIT) Delhi - Department of Computer Science, Title: Neural Circuit Discovery via Representation and Dynamics

November 2026 National Institute for Mathematical and Theoretical Biology, Chicago - Probabilistic Inference in Neuroscience. Title: Inferring Neural Circuits using Diffusion Models